

Ryan Nguyen

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EDUCATION

Bachelor of Computer Science & Mathematics, University of Virginia

Expected May 2027

- GPA: 3.9/4.0
- Relevant courses: Data Structures & Algorithms, Software Development, Computer Systems and Organization

SKILLS

- | | |
|--------------------------|--|
| • Programming Languages | Python, Java, C++, R, SQL, Typescript, JSON, Rust |
| • Frameworks & Libraries | React.js, Next.js, Node.js, TensorFlow, PyTorch, NumPy, MongoDB |
| • Tools & Platforms | Amazon Web Services, Linux, Jupyter Notebook, Anaconda, Git, Vercel, Vim |
| • Miscellaneous | Fusion360, Internet of Things (IoT), Circuit Assembly, Microsoft Office |

EXPERIENCE

StudyStay

Chief Technology Officer

September 2024 – Present

- Cofounded StudyStay to provide a platform for exchange students to find lodging domestically and overseas.
- Spearheaded the development and launch of www.studystay.us to accommodate 500+ unique users in the first month of operation.
- Optimized a **SQL** database and **AWS S3 Bucket** to serve content with <1 second latency, even with complex queries of 100+ images.

UVA Infrastructure Simulation, Sensing and Evaluation Lab (I-S²EE)

Software Engineer Intern

September 2023 – Present

- Authored **STRUCT+**, an **augmented reality app** to perform >80% accuracy structural analysis. See Projects.
- Utilized **machine learning** to simulate structural mechanics without conventional methods or equations.
- Hired as research assistant to the UVA I-S²EE Lab; was selected from a competitive pool of 50+ first- and second-year undergraduates with only one open position.

PROJECTS

Cory: AI Meal Generator Assistant

Personal Project

July 2024 – Present

- Engineered an **AI-powered API** (using **AWS Lambda & API Gateway**) to return curated meal recipes based on user-inputted nutritional values (ex. 15g of fat, 50 g of protein), and return as **JSON** format. 95%+ success rate @ ~3000ms response time.
- Integrated an AWS backend using **DynamoDB** and **S3**. Used **AWS Cognito** for authentication.

STRUCT+: Augmented Reality & ML Structural Analysis

UVA I-S²EE Lab

September 2023 – Present

- Augmented reality**- and **ML**-enabled app to simulate >80% accuracy structural analysis in real time.
- Achieved a <500ms round trip time for invoking the model and displaying visual results back to the screen.
- Coded in **Swift** for the **iOS** platform and used a custom **TensorFlow Lite** model trained from 8000+ inputs.

Personal Resume Website (ryanhnguyen.com)

Personal Project

August 2023 – Present

- Full stack web application with **MongoDB Atlas** deployment in the backend and 3 custom database **APIs**.
- Utilized **React.js** to build and the **Next.js** framework by **Vercel** to deploy. Approximately 8 unique weekly visits.
- Implemented a multi-user reminder system with a real time countdown. Used Google OAuth2 for authentication.

CERTIFICATIONS

Amazon Web Services Certified Cloud Practitioner

Expires June 2027